Anchita Majumdar

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Summary

Detail-oriented B.Tech CSE (AI & ML) graduate (2026) with a strong foundation in Python, SQL and data engineering concepts. Hands-on experience through internships and projects in data cleaning, validation, ETL-style processing and relational databases, including an incremental PySpark and Delta Lake pipeline built on Databricks with MERGE upserts, data quality checks and audit logging. Careful about data quality at every stage of a workflow and comfortable collaborating with teams on workflow design. Eager to build scalable data solutions, learn modern Big Data technologies and contribute to data-driven business applications.

Education

Dr. B. C. Roy Engineering College, Durgapur 2022 – 2026
B.Tech in Computer Science & Engineering (AI & ML) CGPA: 8.34 / 10

Bishnupur High School
Higher Secondary (Class XII), Science Stream Score: 89.2%

Bishnupur Public School
Secondary (Class X) Score: 90.85%

Technical Skills

Data Engineering: Python, SQL, PySpark, Apache Spark, Spark SQL, Delta Lake, ETL / Incremental Loading, Data Cleaning & Validation, Data Quality Checks

Platforms & Tools: Databricks, MySQL, Git, GitHub, Jupyter, Google Colab

Python Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch (CTGAN)

Analytics & Reporting: MS Excel, Power BI, Exploratory Data Analysis, Feature Engineering, Reporting

Machine Learning: Random Forest, K-Means, Model Evaluation, Generative AI, LLMs

Languages: Python, Java

Internships

Data Science Intern, Celebal Technologies | Project Repository Jun 2025 – Aug 2025

- Cleaned, validated and transformed 1,000 loan applicant records (20 variables) using Python and Pandas, engineering features and checking data quality before modelling.
- Built a Random Forest credit-risk model with 80.5% holdout and 76% cross-validation accuracy; analysed precision, recall and F1 by class, finding only 51% of high-risk applicants were caught under a 70/30 imbalance.
- Proposed class weighting, SMOTE and threshold tuning, and recommended routing borderline applicants to manual underwriting using predicted probabilities.
- Documented findings in a structured report to support data-driven decisions.

AI Intern, IEEE CIS | Live Website Jun 2025 – Jul 2025

- Integrated Python-based backend workflows with the CardiaMind AI health system front end (HTML, CSS, JavaScript) for healthcare data analysis and automated reporting.
- Assisted in implementing machine learning models and predictive analysis pipelines, collaborating with the team on workflow design.

Projects

Incremental Orders Data Pipeline (PySpark + Delta Lake) | GitHub
Python, PySpark, Spark SQL, Delta Lake, Databricks, Git

- Built an incremental ETL pipeline on Databricks that loads only new daily order files, using an audit log table to track processed files and make re-runs safe (idempotent).
- Implemented data quality checks (null IDs, invalid amounts, invalid status) that quarantine rejected rows in a separate table, and removed duplicates by keeping the latest record per order with a window function.
- Used Delta Lake `MERGE` to upsert 600 orders across 3 simulated daily loads, updating 60 orders whose status changed without creating duplicates; verified results with Spark SQL checks and Delta table history.
- Documented the architecture, design decisions and run steps in a GitHub README, with work committed in stages.

CTGAN – Synthetic Data Generator | GitHub
Python, CTGAN, PyTorch, Scikit-learn

- Built an end-to-end synthetic tabular data pipeline in Python: label-encoded categorical features, applied an 80/20 train-test split, and trained CTGAN (300 epochs) to generate 500 synthetic rows from a speech dataset.
- Simulated data scarcity by training a Random Forest on 20% of the real training data, then added synthetic rows in steps of 50 (0 to 500); test accuracy rose from 80.8% to 96.2% and F1-score from 0.84 to 0.96.
- Fixed random seeds across NumPy, PyTorch and scikit-learn for reproducible runs, and exported the augmented dataset (real plus synthetic rows) to Excel for downstream reuse.

Trader Performance vs Market Sentiment Analysis | GitHub
Python, Pandas, Scikit-learn, Seaborn

- Processed 200K+ Hyperliquid trade records and joined them with the Bitcoin Fear & Greed Index using a time-based merge, engineering win-rate and daily trader metrics.
- Found average PnL of $206 and a 55% win rate in Extreme Greed versus $1.89 and 29% in Extreme Fear; low-frequency accounts outperformed high-frequency ones in Fear ($246 vs $120 per trade).
- Built a Random Forest model predicting trade profitability (73% accuracy) and clustered accounts into 3 behavioural groups with K-Means.

Website Traffic Analysis & Marketing Insights | GitHub
Python, Pandas, Matplotlib

- Cleaned and aggregated 2,000 website sessions (page views, session duration, bounce rate, time on page, previous visits, traffic source, conversion rate) to evaluate channel performance and user engagement.
- Segmented traffic by source (Organic 786 sessions / 39%, Paid 428, Referral 301, Social 269, Direct 216) and compared conversion by channel; overall conversion was about 98%, so gaps between channels were small.
- Visualised the traffic mix and the bounce-rate vs conversion relationship in Matplotlib, and turned findings into recommendations on channel focus.

Certifications & Achievements

- Software Engineering Job Simulation (Goldman Sachs, 2025)
- Spoken Tutorial Project, IIT Bombay: Java, PHP, MySQL, Linux, LaTeX

Leadership

Secretary, IEEE MTTS Student Branch 2025 – Jan 2026

- Organised technical workshops, coordinated events and managed student engagement activities.